

Lecture 6

systems of nonlinear equations,
Picard's and Newton-Raphson methods for systems

systems of nonlinear equations

Example

Consider

$$\begin{aligned}2x_1 - x_2 + x_1^2 &= \frac{1}{9} \\ -x_1 + 2x_2 + x_2^2 &= \frac{13}{9}\end{aligned}$$

This is a system of two nonlinear equations with two unknowns, x_1 and x_2 . We can always write it in the standard form as:

$$\begin{aligned}2x_1 - x_2 + x_1^2 - \frac{1}{9} &= 0 \\ -x_1 + 2x_2 + x_2^2 - \frac{13}{9} &= 0\end{aligned}$$

Vector notation

Introducing the vectors $\mathbf{x}, \mathbf{0} \in \mathbb{R}^2$ the vectors:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and the vector function $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (input in $\mathbb{R}^2$ and output in $\mathbb{R}^2$):

$$\mathbf{f}(\mathbf{x}) = \mathbf{f}(x_1, x_2) = \begin{bmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{bmatrix} = \begin{bmatrix} 2x_1 - x_2 + x_1^2 - \frac{1}{9} \\ -x_1 + 2x_2 + x_2^2 - \frac{13}{9} \end{bmatrix}$$

the system of nonlinear equation can be written in the *standard* vector equation:

$$\mathbf{f}(\mathbf{x}) = \mathbf{0}$$

or as a vector *fixed-point* problem:

$$\mathbf{g}(\mathbf{x}) = \mathbf{x}$$

For example, with $\mathbf{g}(\mathbf{x}) = \mathbf{f}(\mathbf{x}) + \mathbf{x}$. This notation can be used with any system of m equations with m unknowns, using $\mathbf{x} \in \mathbb{R}^m$ and $\mathbf{f}, \mathbf{g} : \mathbb{R}^m \rightarrow \mathbb{R}^m$.

Basic definitions for vector equations

Vector $\mathbf{p}$ is considered to be a solution of the system $\mathbf{f}(\mathbf{x}) = \mathbf{0}$ if $\mathbf{f}(\mathbf{p}) = \mathbf{0}$.

An *iterative method* produces a sequence of vectors:

$$\mathbf{p}_0, \mathbf{p}_1, \dots$$

The sequence $\mathbf{p}_0, \mathbf{p}_1, \dots$ converges to the vector $\mathbf{p}$ if

$$\lim_{n \rightarrow \infty} \mathbf{p}_n = \mathbf{p}$$

Using a *vector norm* $\|\mathbf{x}\| : \mathbb{R}^m \rightarrow \mathbb{R}^+$ (input in $\mathbb{R}^m$ and output is a positive number), the convergence property can also be expressed as

$$\lim_{n \rightarrow \infty} \|\mathbf{p}_n - \mathbf{p}\| = 0$$

Vector norm and stopping criteria

We denote by p_j , $j = 1, \dots, m$ the elements of the vector $\mathbf{p}$ and by $p_{n,j}$, $j = 1, \dots, m$ the elements of the vector $\mathbf{p}_n$.

An example of a vector norm is the Euclidean norm (also called the 2-norm):

$$\|\mathbf{p}\|_2 = \sqrt{\sum_{j=1}^m p_j^2}, \quad \|\mathbf{p}_n - \mathbf{p}\|_2 = \sqrt{\sum_{j=1}^m (p_{n,j} - p_j)^2}$$

Using vector norms one can also define stopping criteria:

$$\begin{aligned} \|\mathbf{p} - \mathbf{p}_n\| &< \epsilon, && \text{error criterion} \\ \|\mathbf{p}_n - \mathbf{p}_{n-1}\| &< \epsilon, && \text{saturation/stagnation criterion} \\ \|\mathbf{f}(\mathbf{p}_n)\| &< \epsilon, && \text{residual criterion} \end{aligned}$$

Picard's method

Iterations of Picard's method

Picard's method is applied to the fixed-point form of the problem and its iteration looks like:

$$\begin{aligned} \mathbf{f}(\mathbf{x}) = \mathbf{0} &\rightarrow \mathbf{g}(\mathbf{x}) = \mathbf{x} \\ \mathbf{p}_n = \mathbf{g}(\mathbf{p}_{n-1}), \quad n = 1, 2, \dots \end{aligned}$$

Theory of existence, uniqueness and convergence of the Picard method for vector equations is beyond the scope of these Lectures. In short: it does converge (linearly) if the function $\mathbf{g}(\mathbf{x})$ is Lipschitz continuous, has contraction property, and the initial guess $\mathbf{p}_0$ is close enough to the solution $\mathbf{p}$.

Example

$$\begin{aligned}2x_1 - x_2 + x_1^2 &= \frac{1}{9} \\ -x_1 + 2x_2 + x_2^2 &= \frac{13}{9}\end{aligned}$$

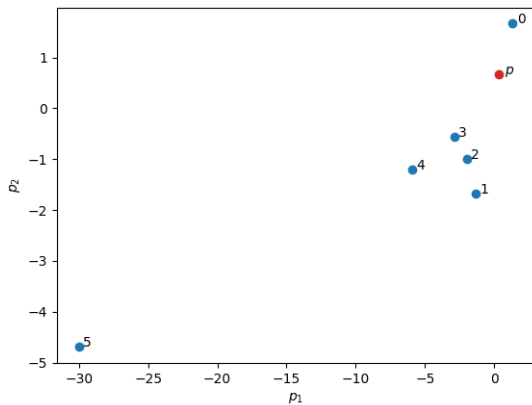
Exact solution $x_1 = 1/3$, $x_2 = 2/3$. Possible fixed-point form:

$$\begin{aligned}x_1 - \left(2x_1 - x_2 + x_1^2 - \frac{1}{9}\right) &= x_1 \\ x_2 - \left(-x_1 + 2x_2 + x_2^2 - \frac{13}{9}\right) &= x_2\end{aligned}$$

Vector form

$$\mathbf{g}(\mathbf{x}) = \begin{bmatrix} -x_1 + x_2 - x_1^2 + \frac{1}{9} \\ x_1 - x_2 - x_2^2 + \frac{13}{9} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{x}$$

Example: Picard's method fails to converge



Example

For the same system

$$\begin{aligned}2x_1 - x_2 + x_1^2 &= \frac{1}{9} \\ -x_1 + 2x_2 + x_2^2 &= \frac{13}{9}\end{aligned}$$

we can derive a fixed-point form with a different $\mathbf{g}(\mathbf{x})$:

$$\begin{aligned}(2 + x_1)x_1 - x_2 &= \frac{1}{9} \\ -x_1 + (2 + x_2)x_2 &= \frac{13}{9}\end{aligned}$$

or

$$\begin{aligned}\begin{bmatrix} 2 + x_1 & -1 \\ -1 & 2 + x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} \frac{1}{9} \\ \frac{13}{9} \end{bmatrix} \\ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 2 + x_1 & -1 \\ -1 & 2 + x_2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{9} \\ \frac{13}{9} \end{bmatrix}\end{aligned}$$

Example

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 + x_1 & -1 \\ -1 & 2 + x_2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{9} \\ \frac{13}{9} \end{bmatrix}$$

This has the vector form:

$$\mathbf{x} = [A(\mathbf{x})]^{-1} \mathbf{b}$$

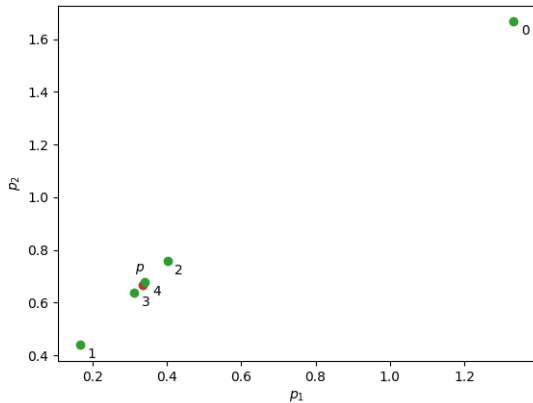
Hence, we can use

$$\mathbf{g}(\mathbf{x}) = [A(\mathbf{x})]^{-1} \mathbf{b}$$

and iterate as:

$$\mathbf{p}_n = [A(\mathbf{p}_{n-1})]^{-1} \mathbf{b}, \quad n = 1, 2, \dots$$

Example: Picard's method converges with alternative $g(x)$



Newton-Raphson method

Recalling the scalar formulation

Let the problem $f(x) = 0$ have the solution $x = p$. Expanding $f(p)$ around $x \neq p$, we get:

$$\begin{aligned}f(p) &= f(x) + f'(x)(p - x) + \frac{f''(\xi)}{2}(p - x)^2, \\0 &= f(x) + f'(x)(p - x) + \frac{f''(\xi)}{2}(p - x)^2,\end{aligned}$$

where ξ is between x and p . Suppose $f'(x) \neq 0$. Then,

$$\begin{aligned}p &= x - \frac{f(x)}{f'(x)} + \frac{f''(\xi)}{2f'(x)}(p - x)^2 \\p &\approx x - \frac{f(x)}{f'(x)}\end{aligned}$$

Iterations:

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n = 1, 2, \dots$$

Taylor expansion for vector functions

Let the problem $\mathbf{f}(\mathbf{x}) = \mathbf{0}$ have the solution $\mathbf{x} = \mathbf{p}$. Expanding the components $f_n(x_1, \dots, x_m)$ of $\mathbf{f}(\mathbf{p})$ around $\mathbf{x} \neq \mathbf{p}$, we get:

$$f_1(\mathbf{p}) = f_1(\mathbf{x}) + \frac{\partial f_1(\mathbf{x})}{\partial x_1}(p_1 - x_1) + \dots + \frac{\partial f_1(\mathbf{x})}{\partial x_m}(p_m - x_m) + \mathcal{O}(\|\mathbf{p} - \mathbf{x}\|^2)$$

...

$$f_m(\mathbf{p}) = f_m(\mathbf{x}) + \frac{\partial f_m(\mathbf{x})}{\partial x_1}(p_1 - x_1) + \dots + \frac{\partial f_m(\mathbf{x})}{\partial x_m}(p_m - x_m) + \mathcal{O}(\|\mathbf{p} - \mathbf{x}\|^2)$$

$$\begin{bmatrix} f_1(\mathbf{p}) \\ \vdots \\ f_m(\mathbf{p}) \end{bmatrix} = \begin{bmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix} + \begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f_1(\mathbf{x})}{\partial x_m} \\ \vdots & \dots & \vdots \\ \frac{\partial f_m(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f_m(\mathbf{x})}{\partial x_m} \end{bmatrix} \begin{bmatrix} p_1 - x_1 \\ \vdots \\ p_m - x_m \end{bmatrix} + \mathcal{O}(\|\mathbf{p} - \mathbf{x}\|^2)$$

$$\mathbf{f}(\mathbf{p}) = \mathbf{f}(\mathbf{x}) + J(\mathbf{x})(\mathbf{p} - \mathbf{x}) + \mathcal{O}(\|\mathbf{p} - \mathbf{x}\|^2)$$

$J(\mathbf{x}) \in \mathbb{R}^{m \times m}$ is called the *Jacobian* of $\mathbf{f}$.

Remainder of the Taylor expansion for vector functions

We have the Taylor expansion in the form:

$$\mathbf{f}(\mathbf{p}) = \mathbf{f}(\mathbf{x}) + J(\mathbf{x})(\mathbf{p} - \mathbf{x}) + \mathcal{O}(\|\mathbf{p} - \mathbf{x}\|^2)$$

For those who are curious about the explicit form of the Lagrange remainder in the vector case, it looks like this:

$$f_i(\mathbf{p}) = f_i(\mathbf{x}) + \sum_{j=1}^m \frac{\partial f_i(\mathbf{x})}{\partial x_j} (p_j - x_j) + \frac{1}{2} \sum_{j=1}^m \sum_{k=1}^m \frac{\partial^2 f_i(\boldsymbol{\xi})}{\partial x_j \partial x_k} (p_j - x_j)(p_k - x_k)$$

where $\boldsymbol{\xi}$ is situated somewhere on the line (in m -dimensional space) connecting $\mathbf{p}$ and $\mathbf{x}$.

The term with all second partial derivatives of all components of $\mathbf{f}$ is a tensor of rank three:

$$H_{i,j,k}(\mathbf{x}) = \frac{\partial^2 f_i(\mathbf{x})}{\partial x_j \partial x_k}, \quad i = 1, \dots, m; j = 1, \dots, m; k = 1, \dots, m;$$

and is called the *Hessian* of $\mathbf{f}$.

Vector formulation of the Newton-Raphson method

Since $\mathbf{f}(\mathbf{p}) = \mathbf{0}$,

$$\mathbf{0} = \mathbf{f}(\mathbf{x}) + J(\mathbf{x})(\mathbf{p} - \mathbf{x}) + \mathcal{O}(\|\mathbf{p} - \mathbf{x}\|^2)$$

Suppose $J(\mathbf{x})$ is an *invertible* matrix. Then,

$$\mathbf{p} = \mathbf{x} - [J(\mathbf{x})]^{-1}\mathbf{f}(\mathbf{x}) + \mathcal{O}(\|\mathbf{p} - \mathbf{x}\|^2)$$

$$\mathbf{p} \approx \mathbf{x} - [J(\mathbf{x})]^{-1}\mathbf{f}(\mathbf{x})$$

Iterations:

$$\mathbf{p}_n = \mathbf{p}_{n-1} - [J(\mathbf{p}_{n-1})]^{-1}\mathbf{f}(\mathbf{p}_{n-1}), \quad n = 1, 2, \dots$$

Computationally more efficient way:

$$J(\mathbf{p}_{n-1})\mathbf{v}_{n-1} = \mathbf{f}(\mathbf{p}_{n-1}), \quad \text{solve this linear system for } \mathbf{v}_{n-1}$$

$$\mathbf{p}_n = \mathbf{p}_{n-1} - \mathbf{v}_{n-1}, \quad \text{update}$$

$$n = 1, 2, \dots$$

Example

Again the system

$$\begin{aligned}2x_1 - x_2 + x_1^2 &= \frac{1}{9} \\ -x_1 + 2x_2 + x_2^2 &= \frac{13}{9}\end{aligned}$$

In the standard vector form $\mathbf{f}(\mathbf{x}) = \mathbf{0}$:

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} 2x_1 + x_1^2 - x_2 - \frac{1}{9} \\ -x_1 + 2x_2 + x_2^2 - \frac{13}{9} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{0}$$

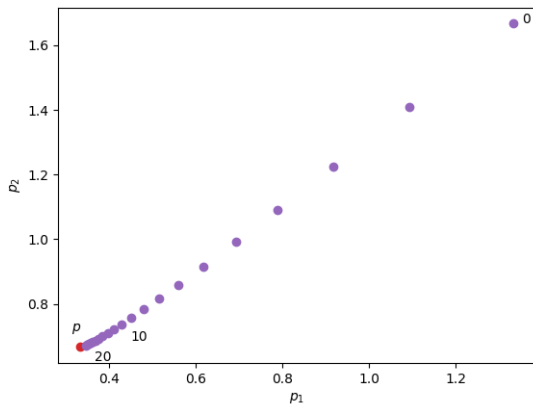
Hence,

$$\begin{aligned}f_1(\mathbf{x}) &= 2x_1 + x_1^2 - x_2 - \frac{1}{9} \\ f_2(\mathbf{x}) &= -x_1 + 2x_2 + x_2^2 - \frac{13}{9}\end{aligned}$$

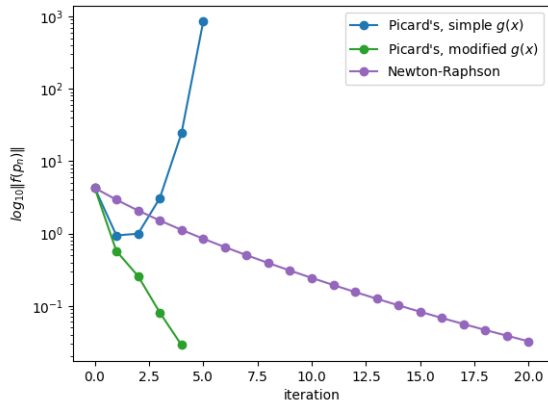
Computing the Jacobian:

$$J(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2 + 2x_1 & -1 \\ -1 & 2 + 2x_2 \end{bmatrix}$$

Newton-Raphson iterations: $\mathbf{p}_n = \mathbf{p}_{n-1} - [J(\mathbf{p}_{n-1})]^{-1}\mathbf{f}(\mathbf{p}_{n-1})$



Convergence plots: norm of the residual $\|f(p_n)\|$



Summary: systems of nonlinear algebraic equations

- Picard's method is easy to apply, but may fail to converge
- Alternative fixed-point forms of the problem may lead to convergent Picard's iterations
- Newton-Raphson method is more robust and often converges without problem reformulation
- Newton-Raphson method requires updating the Jacobian matrix and solving a new linear system at each iteration
- If Jacobian matrix is not available, then its finite-difference approximation may be used

Questions

- Given a system of two nonlinear equations with two unknowns and some initial guess, you should be able to write down one iteration of the Picard method (any choice of $\mathbf{g}(\mathbf{x})$) and one iteration of the Newton-Raphson method, including explicit calculations of $J(\mathbf{x})$ and $[J(\mathbf{x})]^{-1}$.